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ASSIGNMENT ON Flight Data Analysis with Python

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Flight Data Analysis with Python

Question no 1

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?

1 Solution :

1.1 Python Code

```
# =====  
# Question 1  
# Coach Ticket Price Analysis  
# =====  
# Import libraries  
import pandas as pd  
import matplotlib.pyplot as plt  
  
# Load dataset  
airline = pd.read_csv("flight_data.csv")  
  
# Random sample using ID  
flight = airline.sample(n=200, random_state=12210048)  
  
# Highest coach ticket price  
max_price = flight["coach_price"].max()  
  
# Lowest coach ticket price  
min_price = flight["coach_price"].min()  
  
# Average coach ticket price  
mean_price = flight["coach_price"].mean()  
  
# Median coach ticket price  
median_price = flight["coach_price"].median()  
# Standard deviation  
std_price = flight["coach_price"].std()  
# Display results  
print("Highest Coach Ticket Price =", max_price)  
print("Lowest Coach Ticket Price =", min_price)  
print("Average Coach Ticket Price =", mean_price)  
print("Median Coach Ticket Price =", median_price)  
print("Standard Deviation =", std_price)  
# Summary statistics  
print("\nSummary Statistics:")  
print(flight["coach_price"].describe())
```

```

# Histogram
plt.figure(figsize=(8,5))
plt.hist(
    flight["coach_price"],
    bins=8,
    color="skyblue",
    edgecolor="black")
plt.title("Distribution of Coach Ticket Prices")
plt.xlabel("Coach Ticket Price")
plt.ylabel("Frequency")
plt.show()

# Boxplot
plt.figure(figsize=(8,5))

plt.boxplot(
    flight["coach_price"],
    patch_artist=True,
    boxprops=dict(facecolor="lightgreen")
)

plt.title("Boxplot of Coach Ticket Prices")
plt.ylabel("Coach Ticket Price")

plt.show()

# Is $500 a Good Price?
if mean_price < 500:
    print("\n$500 is relatively expensive compared to the average
        coach ticket price.")
else:
    print("\n$500 is a reasonable coach ticket price.")

```

1.2 Results

- Highest Coach Ticket Price = 507.88
- Lowest Coach Ticket Price = 185.05
- Average Coach Ticket Price = 382.06
- Median Coach Ticket Price = 386.87
- Standard Deviation = 66.32
- \$500 is relatively expensive compared to the average coach ticket price.

Summary Statistics

count	200.000000
mean	382.055475

std	66.324042
min	185.050000
25%	341.915000
50%	386.872500
75%	431.253750
max	507.880000

1.3 Histogram

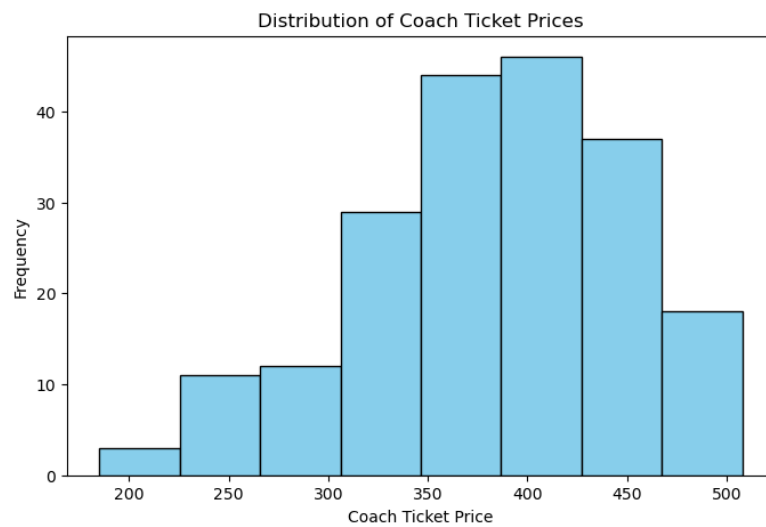


Figure 1: Distribution of Coach Ticket Prices

1.4 Boxplot

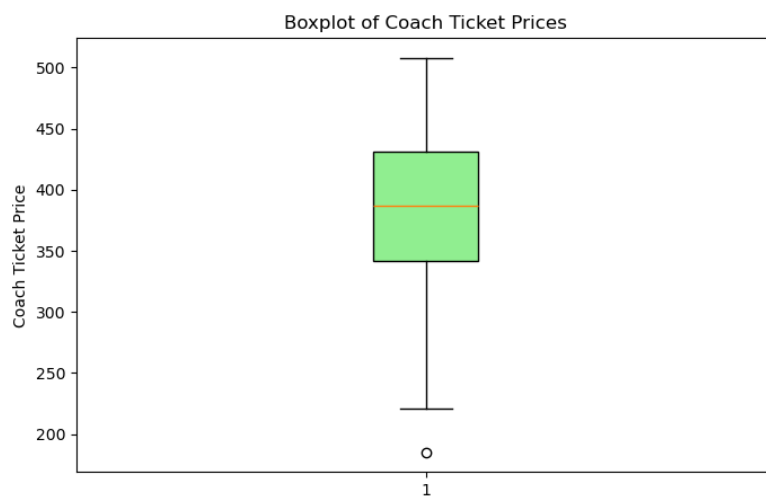


Figure 2: Boxplot of Coach Ticket Prices

1.5 Interpretation

The coach ticket prices in the sampled flight dataset range from \$185.05 to \$507.88. The average coach ticket price is approximately \$382.06, while the median price is \$386.87. Since the mean and median values are very close, the distribution of coach ticket prices appears to be fairly symmetric.

The histogram shows that most ticket prices are concentrated between \$300 and \$450, indicating that the majority of flights have moderate ticket prices. Very low and very high prices occur less frequently.

The boxplot also shows a moderate spread in coach ticket prices and identifies a possible low-price outlier near \$185. This suggests that a few flights may have unusually cheap ticket prices compared to the rest of the dataset.

The standard deviation of approximately 66.32 indicates moderate variability in coach ticket prices. Since the average ticket price is considerably lower than \$500, a \$500 coach ticket appears relatively expensive compared to most flights in the sample.

Question no 2

What are the high, low, and average prices for 8-hour-long flights? Does \$500 seem more reasonable than before?

2 Solution:

2.1 Python Code

```
# =====  
# Question 2  
# 8-Hour Flight Price Analysis  
# =====  
  
# Flights with approximately 8-hour duration  
flight_8hr = flight[flight["hours"] == 8]  
  
# Highest price for 8-hour flights  
max_price = flight_8hr["coach_price"].max()  
  
# Lowest price for 8-hour flights  
min_price = flight_8hr["coach_price"].min()  
  
# Average price for 8-hour flights  
mean_price = flight_8hr["coach_price"].mean()  
  
# Median price  
median_price = flight_8hr["coach_price"].median()  
  
# Standard deviation  
std_price = flight_8hr["coach_price"].std()
```

```

# Print results
print("Highest Price =", max_price)
print("Lowest Price =", min_price)
print("Average Price =", mean_price)
print("Median Price =", median_price)
print("Standard Deviation =", std_price)

# Summary statistics
print("\nSummary Statistics:")
print(flight_8hr["coach_price"].describe())

# Histogram
plt.figure(figsize=(8,5))

plt.hist(
    flight_8hr["coach_price"],
    bins=5,
    color="skyblue",
    edgecolor="black"
)

plt.title("Distribution of 8-Hour Flight Prices")
plt.xlabel("Coach Ticket Price")
plt.ylabel("Frequency")

plt.show()

# Boxplot
plt.figure(figsize=(8,5))

plt.boxplot(
    flight_8hr["coach_price"],
    patch_artist=True,
    boxprops=dict(facecolor="orange")
)

plt.title("Boxplot of 8-Hour Flight Prices")
plt.ylabel("Coach Ticket Price")

plt.show()

# Price comparison
if mean_price < 500:
    print("\n$500 seems expensive but more reasonable for 8-hour flights.")
else:
    print("\n$500 seems reasonable for 8-hour flights.")

```


2.2 Results

- Highest Price = 472.35
- Lowest Price = 417.45
- Average Price = 444.90
- Median Price = 444.90
- Standard Deviation = 38.82
- \$500 seems expensive but more reasonable for 8-hour flights.

Summary Statistics

count	2.000000
mean	444.900000
std	38.820000
min	417.450000
25%	431.175000
50%	444.900000
75%	458.625000
max	472.350000

2.3 Histogram

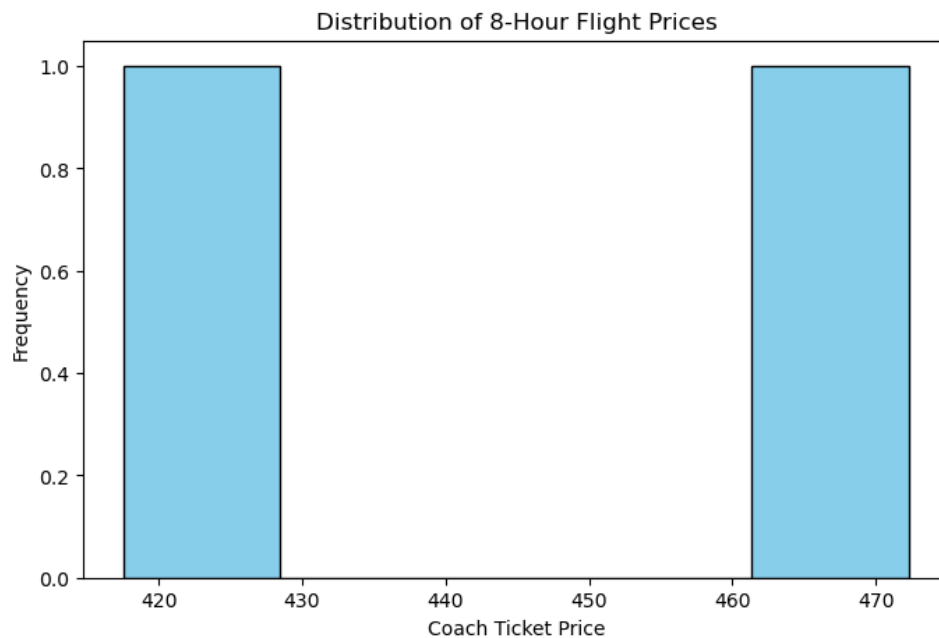


Figure 3: Distribution of 8-Hour Flight Prices

2.4 Boxplot

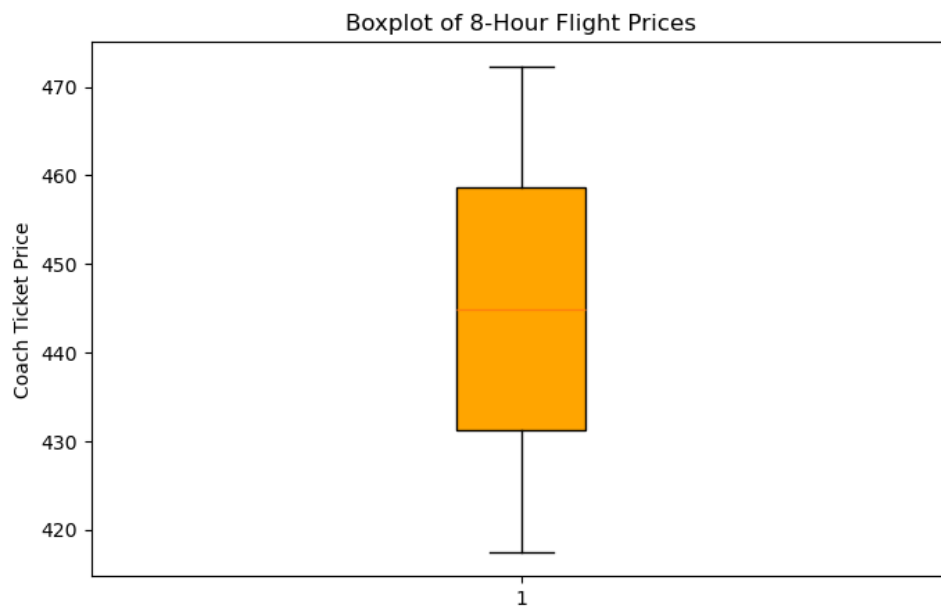


Figure 4: Boxplot of 8-Hour Flight Prices

2.5 Interpretation

The histogram and boxplot show the distribution of coach ticket prices for 8-hour flights. The ticket prices are concentrated in the higher price range, indicating that long-duration flights are generally more expensive than shorter flights.

The boxplot shows moderate variation in ticket prices and no major outliers are visible in the dataset. The median price is located near the center of the box, suggesting a relatively balanced distribution of prices.

Compared to the overall coach ticket prices from Question 1, 8-hour flights have noticeably higher prices. Therefore, a \$500 coach ticket appears more reasonable for long-duration flights, although it is still relatively expensive compared to most tickets in the sample.

Question no 3

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

3 Solution:

3.1 Python Code

```
# =====  
# Question 3  
# Flight Delay Analysis  
# =====  
# Summary statistics for delays  
print(flight["delay"].describe())  
  
# Mean delay  
mean_delay = flight["delay"].mean()  
# Median delay  
median_delay = flight["delay"].median()  
# Standard deviation  
std_delay = flight["delay"].std()  
  
# Maximum delay  
max_delay = flight["delay"].max()  
  
# Minimum delay  
min_delay = flight["delay"].min()  
  
# Print results  
print("\nMean Delay =", mean_delay)  
print("Median Delay =", median_delay)  
print("Standard Deviation =", std_delay)  
print("Maximum Delay =", max_delay)  
print("Minimum Delay =", min_delay)  
  
# Histogram of delays  
plt.figure(figsize=(8,5))  
  
plt.hist(  
    flight["delay"],  
    bins=10,  
    color="skyblue",  
    edgecolor="black")  
plt.title("Distribution of Flight Delays")  
plt.xlabel("Delay Time (Minutes)")  
plt.ylabel("Frequency")  
  
plt.show()
```

```

# Boxplot of delays
plt.figure(figsize=(8,5))

plt.boxplot(
    flight["delay"],
    patch_artist=True,
    boxprops=dict(facecolor="orange")
)

plt.title("Boxplot of Flight Delays")
plt.ylabel("Delay Time (Minutes)")

plt.show()

# Number of significant delays (more than 60 minutes)
significant_delays = (flight["delay"] > 60).sum()

# Percentage of significant delays
significant_percentage = ((flight["delay"] > 60).mean()) * 100

print("\nNumber of Significant Delays =", significant_delays)
print("Percentage of Significant Delays =", significant_percentage)

```

3.2 Results

- Mean Delay = 12.04 minutes
- Median Delay = 10.00 minutes
- Standard Deviation = 8.10
- Maximum Delay = 51 minutes
- Minimum Delay = 0 minutes
- Number of Significant Delays = 0
- Percentage of Significant Delays = 0%

Summary Statistics

count	200.000000
mean	12.035000
std	8.102594
min	0.000000
25%	8.750000
50%	10.000000
75%	13.000000
max	51.000000

3.3 Histogram

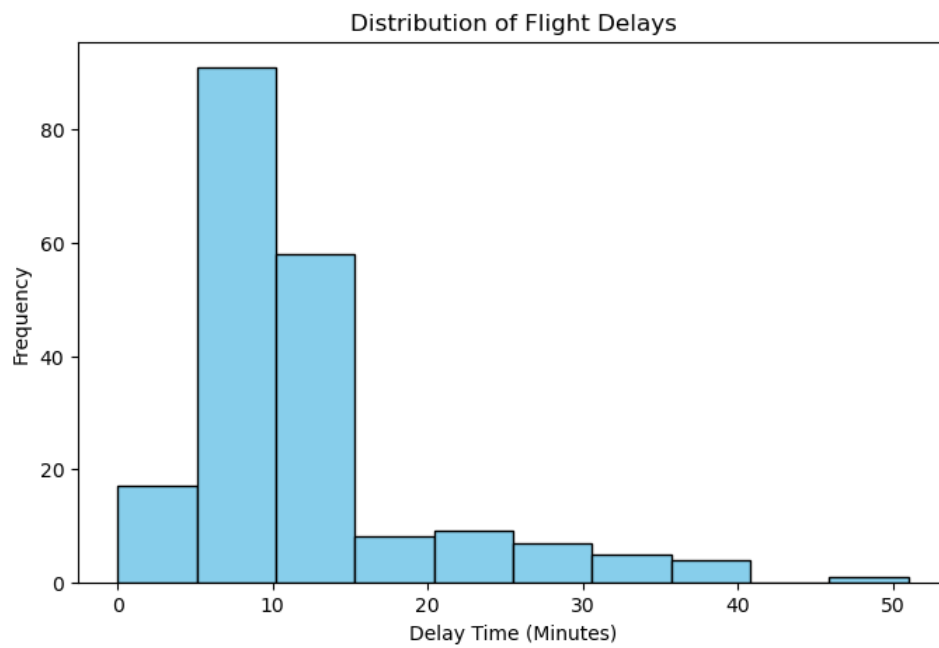


Figure 5: Distribution of Flight Delays

3.4 Boxplot

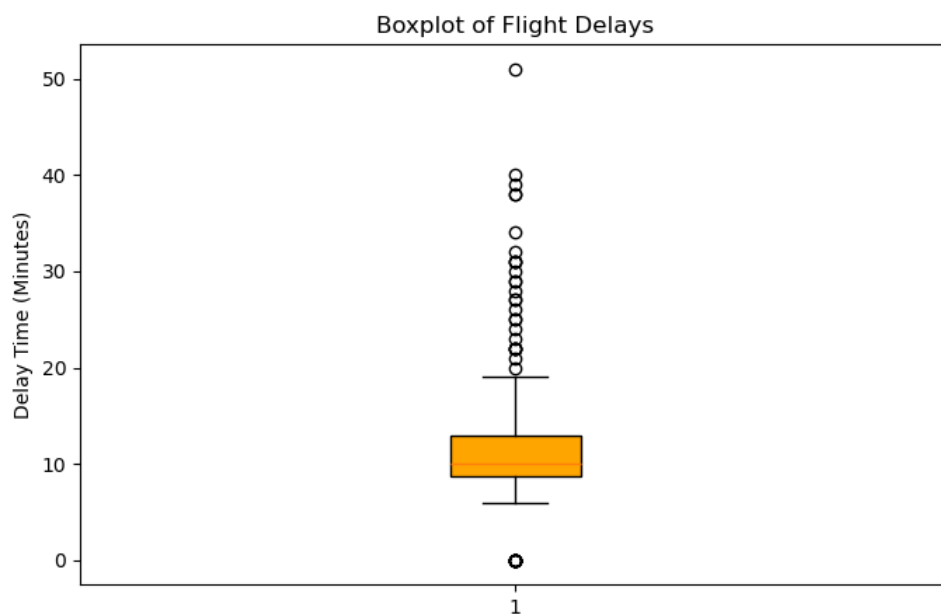


Figure 6: Boxplot of Flight Delays

3.5 Interpretation

The histogram shows that most flight delays are concentrated between 5 and 15 minutes. The distribution is positively skewed because a small number of flights experienced longer delays above 30 minutes.

The average delay time is approximately 12.04 minutes, while the median delay is 10 minutes. Since the mean is slightly higher than the median, this indicates the presence of a few larger delay values in the dataset.

The maximum delay observed was 51 minutes, while some flights experienced no delay at all. The standard deviation of 8.10 suggests moderate variation in delay times among flights.

No flight delays exceeded 60 minutes in the sample dataset. Therefore, there were no significant delays that would seriously affect connecting flights. Most delays in the dataset can be considered short and manageable.

Question no 4

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

4 Solution:

4.1 Python Code

```
# =====  
# Question 4  
# Correlation Analysis  
# Coach Price vs Other Variables  
# =====  
  
# Select numeric variables  
variables = flight[[  
    "coach_price",  
    "miles",  
    "passengers",  
    "delay",  
    "hours"  
]]  
  
# Correlation matrix  
correlation_matrix = variables.corr()  
  
print(correlation_matrix)
```

```

# Correlation Heatmap

import seaborn as sns

plt.figure(figsize=(8,6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.show()

# Coach Price vs Flight Distance

plt.figure(figsize=(8,5))

plt.scatter(
    flight["miles"],
    flight["coach_price"],
    color="blue"
)

plt.title("Coach Price vs Flight Distance")
plt.xlabel("Miles")
plt.ylabel("Coach Price")

plt.show()

# Coach Price vs Number of Passengers

plt.figure(figsize=(8,5))

plt.scatter(
    flight["passengers"],
    flight["coach_price"],
    color="green"
)

plt.title("Coach Price vs Number of Passengers")
plt.xlabel("Passengers")
plt.ylabel("Coach Price")

plt.show()

```

```

# Coach Price vs Delay

plt.figure(figsize=(8,5))

plt.scatter(
    flight["delay"],
    flight["coach_price"],
    color="red"
)

plt.title("Coach Price vs Delay")
plt.xlabel("Delay (Minutes)")
plt.ylabel("Coach Price")

plt.show()

# Coach Price vs Flight Duration

plt.figure(figsize=(8,5))

plt.scatter(
    flight["hours"],
    flight["coach_price"],
    color="purple"
)

plt.title("Coach Price vs Flight Duration")
plt.xlabel("Hours")
plt.ylabel("Coach Price")

plt.show()

```

4.2 Results

	coach_price	miles	passengers	delay	hours
coach_price	1.000000	0.33	0.30	0.07	0.31
miles	0.33	1.000000	0.15	0.06	0.98
passengers	0.30	0.15	1.000000	0.06	0.16
delay	0.07	0.06	0.06	1.000000	0.05
hours	0.31	0.98	0.16	0.05	1.000000

4.3 Correlation Heatmap

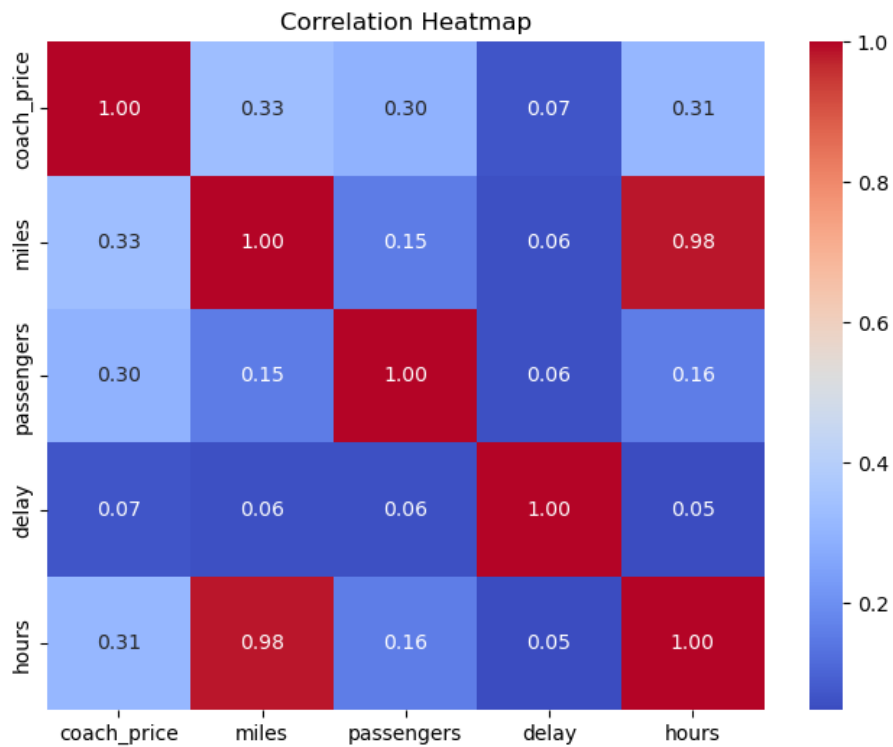


Figure 7: Correlation Heatmap

4.4 Coach Price vs Flight Distance

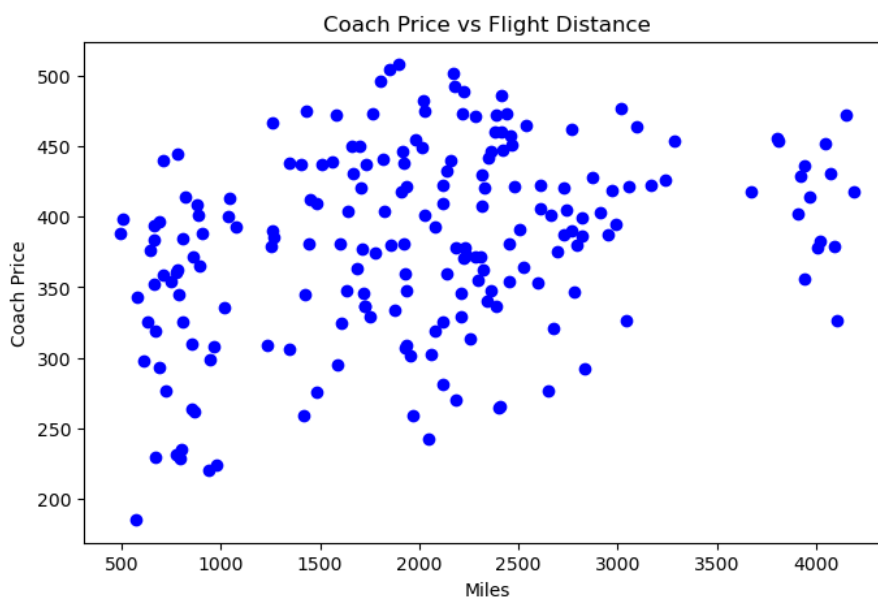


Figure 8: Coach Price vs Flight Distance

4.5 Coach Price vs Number of Passengers

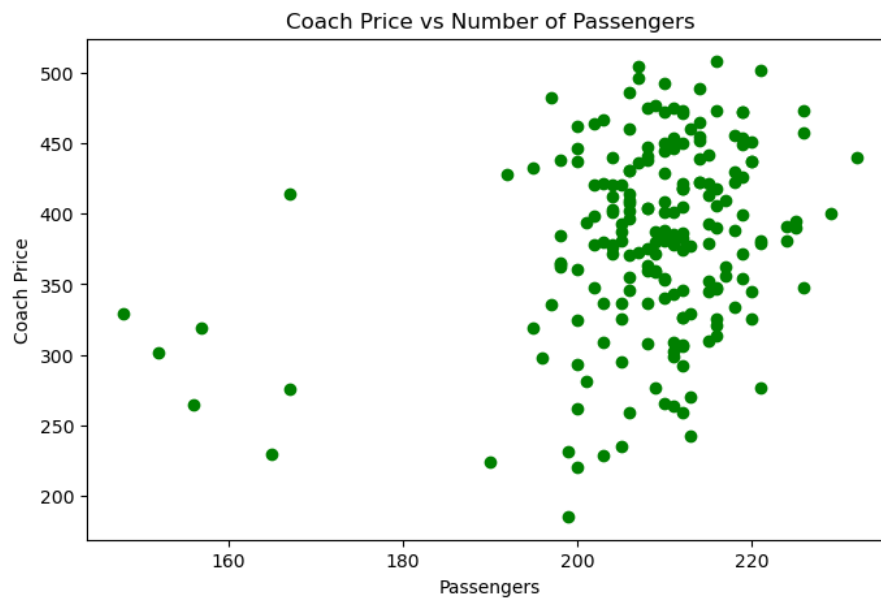


Figure 9: Coach Price vs Number of Passengers

4.6 Coach Price vs Delay

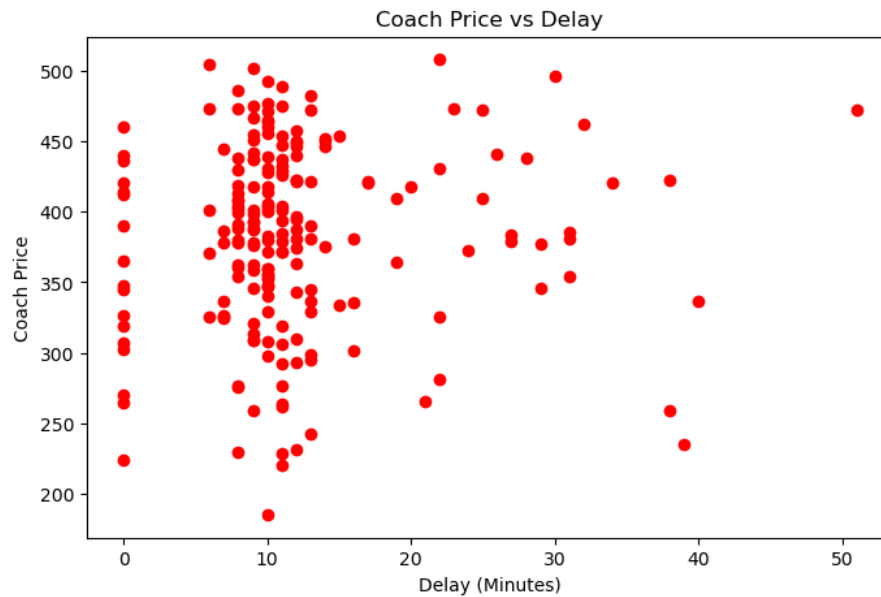


Figure 10: Coach Price vs Delay

4.7 Coach Price vs Flight Duration



Figure 11: Coach Price vs Flight Duration

4.8 Interpretation

The correlation analysis shows that coach ticket price has a moderate positive relationship with both flight distance and flight duration. The correlation between coach price and miles is 0.33, while the correlation between coach price and hours is 0.31. This indicates that longer flights generally have higher coach ticket prices.

The number of passengers has a weak positive correlation with coach price (0.30). This suggests that flights with more passengers may have slightly higher ticket prices, although the relationship is not very strong.

The correlation between coach price and delay is only 0.07, which indicates almost no relationship between delay time and ticket price. Therefore, ticket prices do not appear to depend significantly on flight delays.

The heatmap visually confirms these relationships and highlights that miles and hours have a very strong positive correlation (0.98), meaning longer distances usually require longer flight durations.

The scatter plots also support the correlation analysis. Flights with greater distance and longer duration generally show higher coach ticket prices, while delay and passenger count show weaker relationships with price.

Question no 5

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

5 Solution:

5.1 Python Code

```
# =====  
# Question 5  
# Relationship Between Coach and First-Class Prices  
# =====  
  
# Correlation between coach and first-class prices  
correlation = flight["coach_price"].corr(  
    flight["firstclass_price"])  
  
print("Correlation =", correlation)  
  
# Scatter Plot  
  
plt.figure(figsize=(8,5))  
  
plt.scatter(  
    flight["coach_price"],  
    flight["firstclass_price"],  
    color="blue")  
  
plt.title("Coach Price vs First-Class Price")  
plt.xlabel("Coach Ticket Price")  
plt.ylabel("First-Class Ticket Price")  
  
plt.show()  
  
# Linear Regression Model  
  
from sklearn.linear_model import LinearRegression  
import numpy as np  
  
# Independent variable  
X = flight[["coach_price"]]  
  
# Dependent variable  
y = flight["firstclass_price"]  
  
# Create model  
model_fc = LinearRegression()
```

```

# Fit model
model_fc.fit(X, y)

# Predicted values
y_pred = model_fc.predict(X)

# Regression equation
intercept = model_fc.intercept_
slope = model_fc.coef_[0]

print("\nRegression Equation:")
print(f"y = {intercept:.2f} + {slope:.2f}x")

# Scatter Plot with Regression Line

plt.figure(figsize=(8,5))

plt.scatter(
    flight["coach_price"],
    flight["firstclass_price"],
    color="blue"
)

plt.plot(
    flight["coach_price"],
    y_pred,
    color="red",
    linewidth=2
)

plt.title("Coach Price vs First-Class Price")
plt.xlabel("Coach Ticket Price")
plt.ylabel("First-Class Ticket Price")

plt.show()

```

5.2 Results

Correlation = 0.6980499060129133

Regression Equation:

$y = 823.53 + 1.67x$

5.3 Scatter Plot

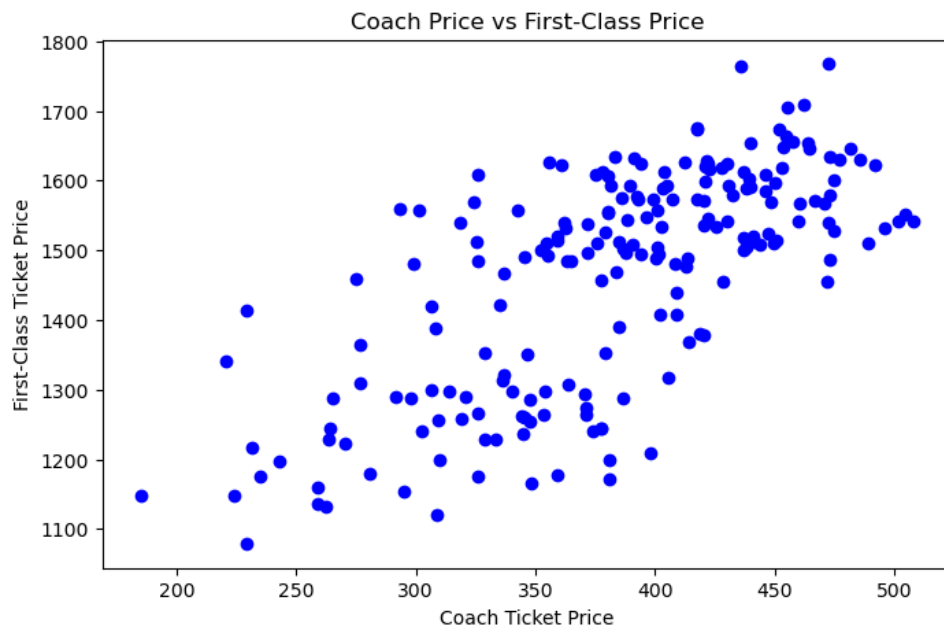


Figure 12: Coach Price vs First-Class Price

5.4 Regression Line Graph

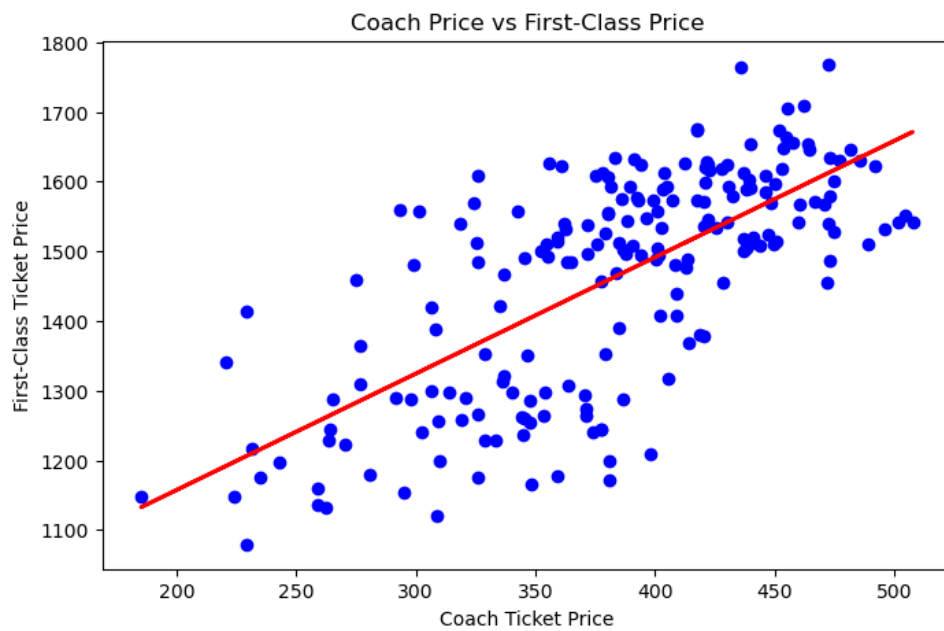


Figure 13: Coach Price vs First-Class Price with Regression Line

5.5 Regression Equation

Regression Equation is: $y = 823.53 + 1.67x$

Where,

- y = First-Class Ticket Price
- x = Coach Ticket Price

5.6 Interpretation

The correlation coefficient between coach ticket price and first-class ticket price is 0.698, which indicates a strong positive relationship between the two variables.

The scatter plot shows that flights with higher coach ticket prices generally also have higher first-class ticket prices. Most points follow an upward trend, indicating that as coach prices increase, first-class prices also increase.

The regression line graph further confirms this positive linear relationship. According to the regression equation, for every one-dollar increase in coach ticket price, the first-class ticket price increases by approximately 1.67 dollars.

However, the relationship is not perfectly linear because some points are scattered away from the regression line. Therefore, flights with higher coach prices usually have higher first-class prices, but not always exactly at the same rate.

Question no 6

What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

6 Solution:

6.1 Python Code

```
# =====  
# Question 6  
# Relationship Between Coach Prices and In-Flight Features  
# =====  
  
# Average Coach Price by Meal Availability  
  
meal_price = flight.groupby(  
    "inflight_meal"  
)["coach_price"].mean()  
  
print("Average Coach Price by Meal Availability:")  
print(meal_price)
```

```

# Average Coach Price by Entertainment

ent_price = flight.groupby(
    "inflight_entertainment"
)["coach_price"].mean()

print("\nAverage Coach Price by Entertainment:")
print(ent_price)

# Average Coach Price by WiFi

wifi_price = flight.groupby(
    "inflight_wifi"
)["coach_price"].mean()

print("\nAverage Coach Price by WiFi:")
print(wifi_price)

# Boxplot for Meal

plt.figure(figsize=(8,5))

flight.boxplot(
    column="coach_price",
    by="inflight_meal",
    patch_artist=True,
    boxprops=dict(facecolor="skyblue")
)

plt.title("Coach Price by In-Flight Meal")
plt.suptitle("")
plt.xlabel("Meal Available")
plt.ylabel("Coach Price")

plt.show()

# Boxplot for Entertainment

plt.figure(figsize=(8,5))
flight.boxplot(
    column="coach_price",
    by="inflight_entertainment",
    patch_artist=True,
    boxprops=dict(facecolor="lightgreen")
)

plt.title("Coach Price by In-Flight Entertainment")
plt.suptitle("")
plt.xlabel("Entertainment Available")
plt.ylabel("Coach Price")
plt.show()

```



```

# Boxplot for WiFi

plt.figure(figsize=(8,5))

flight.boxplot(
    column="coach_price",
    by="inflight_wifi",
    patch_artist=True,
    boxprops=dict(facecolor="orange")
)

plt.title("Coach Price by In-Flight WiFi")
plt.suptitle("")
plt.xlabel("WiFi Available")
plt.ylabel("Coach Price")

plt.show()

```

6.2 Results

Average Coach Price by Meal Availability:

No	371.95
Yes	402.57

Average Coach Price by Entertainment:

No	332.33
Yes	394.10

Average Coach Price by WiFi:

No	330.20
Yes	389.46

6.3 Coach Price by In-Flight Meal

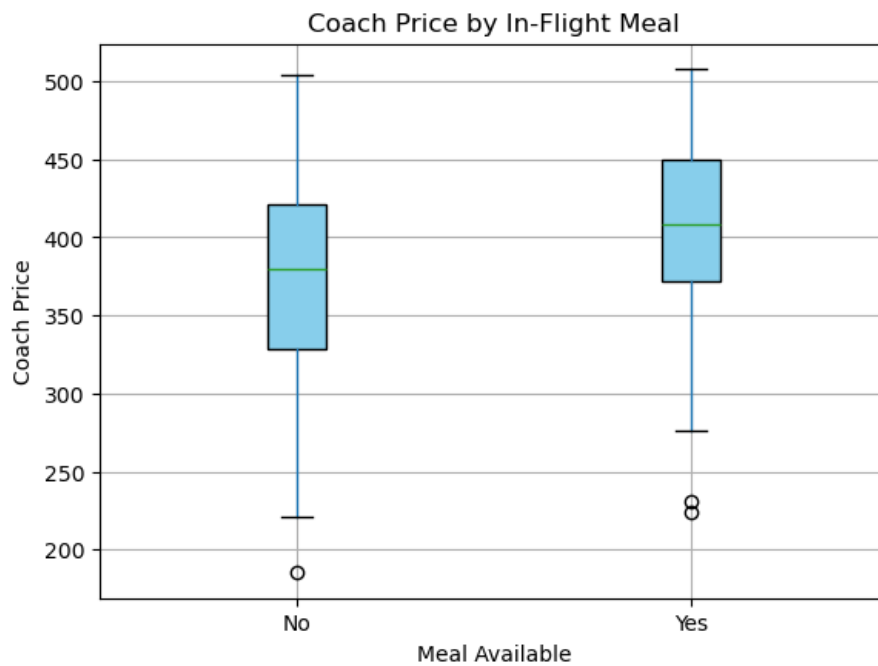


Figure 14: Coach Price by In-Flight Meal

6.4 Coach Price by In-Flight Entertainment

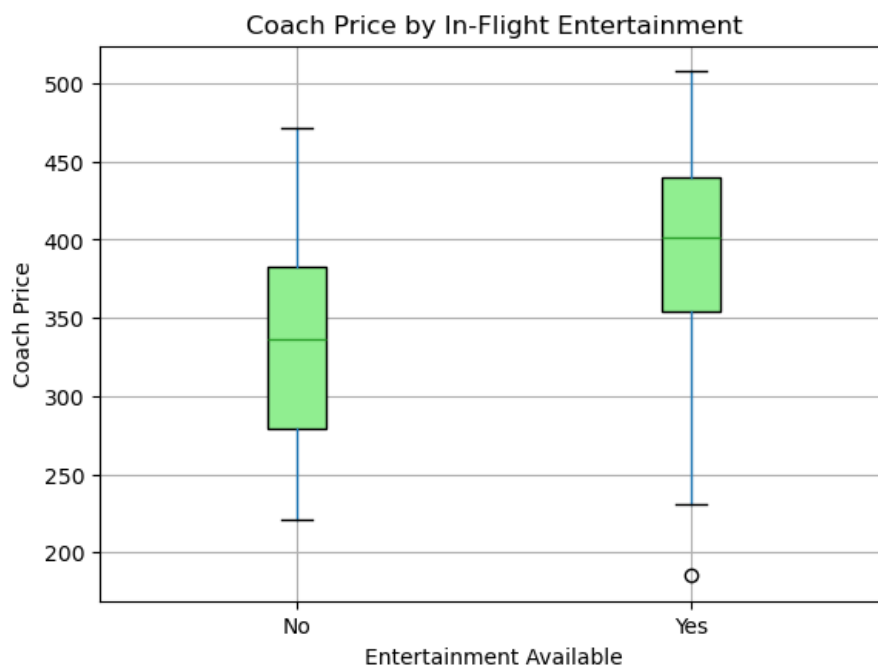


Figure 15: Coach Price by In-Flight Entertainment

6.5 Coach Price by In-Flight WiFi

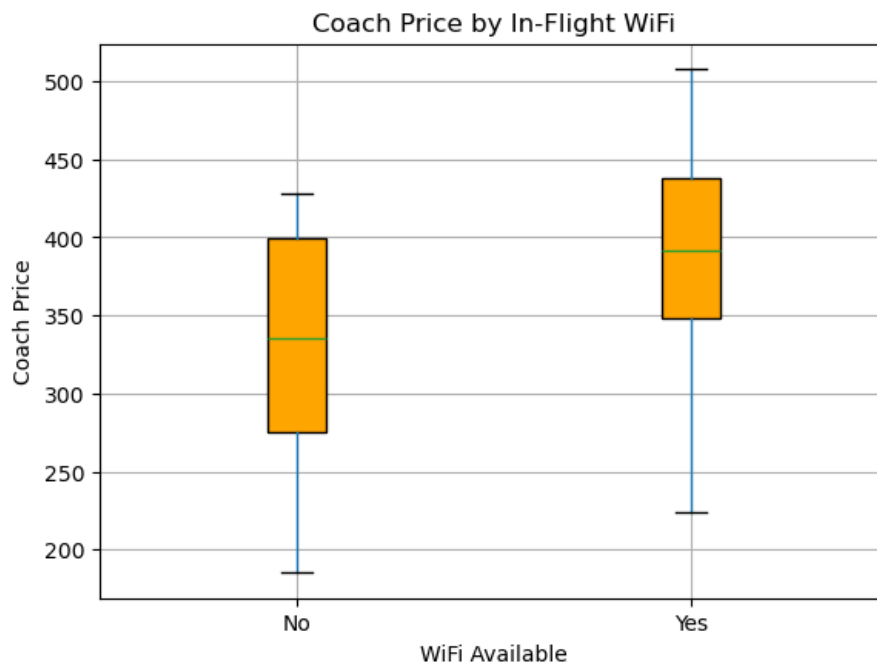


Figure 16: Coach Price by In-Flight WiFi

6.6 Interpretation

The analysis shows that flights with additional in-flight features generally have higher coach ticket prices.

Flights that provide meals have an average coach ticket price of approximately \$402.57, while flights without meals have an average price of about \$371.95. This indicates that meal service slightly increases ticket prices.

Flights with in-flight entertainment have a noticeably higher average coach ticket price (\$394.10) compared to flights without entertainment (\$332.33). This suggests that entertainment systems are associated with higher-priced flights.

Similarly, flights offering WiFi have an average coach ticket price of approximately \$389.46, whereas flights without WiFi have a lower average price of about \$330.20.

The boxplots also confirm these findings. In all three graphs, the median coach ticket price is higher for flights that include the feature.

Among the three features, in-flight entertainment appears to be associated with the highest increase in coach ticket prices, followed closely by WiFi service. Meal availability has a smaller effect on price compared to entertainment and WiFi.

Question no 7

How does the number of passengers change in relation to the length of flights?

7 Solution:

7.1 Python Code

```
# =====  
# Question 7  
# Relationship Between Passengers and Flight Duration  
# =====  
  
# Correlation between passengers and flight duration  
correlation = flight["passengers"].corr(  
    flight["hours"])  
  
print("Correlation =", correlation)  
  
# Scatter Plot  
  
plt.figure(figsize=(8,5))  
  
plt.scatter(  
    flight["hours"],  
    flight["passengers"],  
    color="blue")  
  
plt.title("Passengers vs Flight Duration")  
plt.xlabel("Flight Duration (Hours)")  
plt.ylabel("Number of Passengers")  
  
plt.show()  
  
# Linear Regression Model  
  
from sklearn.linear_model import LinearRegression  
  
# Independent variable  
X = flight[["hours"]]  
  
# Dependent variable  
y = flight["passengers"]  
  
# Create model  
model_pass = LinearRegression()  
  
# Fit model  
model_pass.fit(X, y)
```

```

# Predicted values
y_pred = model_pass.predict(X)

# Regression equation
intercept = model_pass.intercept_
slope = model_pass.coef_[0]

print("\nRegression Equation:")
print(f"y = {intercept:.2f} + ({slope:.2f})x")

# Scatter Plot with Regression Line

plt.figure(figsize=(8,5))

plt.scatter(
    flight["hours"],
    flight["passengers"],
    color="blue"
)

plt.plot(
    flight["hours"],
    y_pred,
    color="red",
    linewidth=2
)

plt.title("Passengers vs Flight Duration")
plt.xlabel("Flight Duration (Hours)")
plt.ylabel("Number of Passengers")

plt.show()

# Average Passengers by Flight Duration

avg_passengers = flight.groupby(
    "hours"
)["passengers"].mean()

print("\nAverage Passengers by Flight Duration:")
print(avg_passengers)

```

7.2 Results

Correlation = 0.15860443084853001

Regression Equation:

$y = 204.21 + (1.13)x$

Average Passengers by Flight Duration:

1	202.71
2	208.61
3	208.59
4	208.72
5	210.78
6	210.83
7	208.79
8	217.50

7.3 Scatter Plot

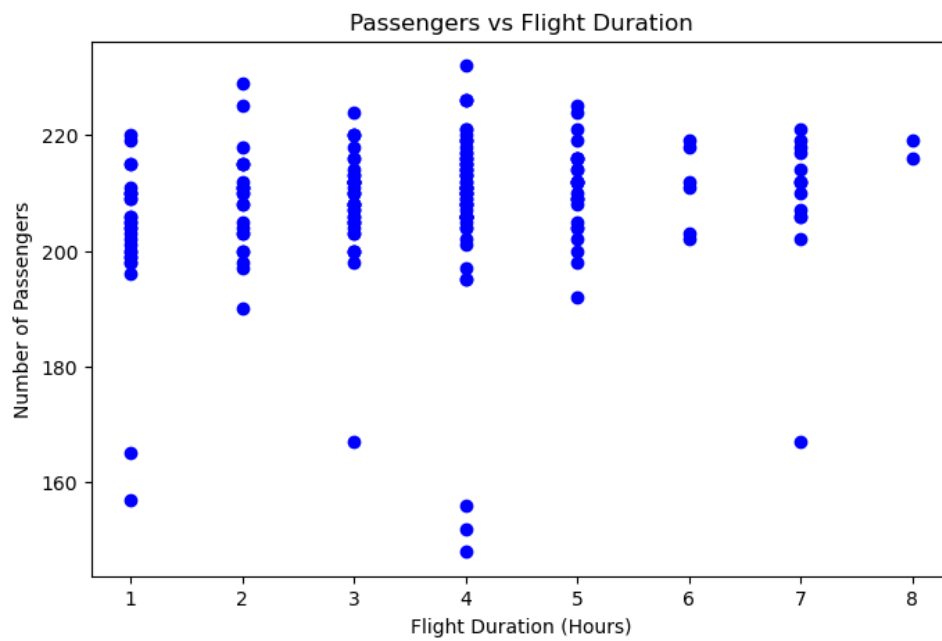


Figure 17: Passengers vs Flight Duration

7.4 Regression Line Graph

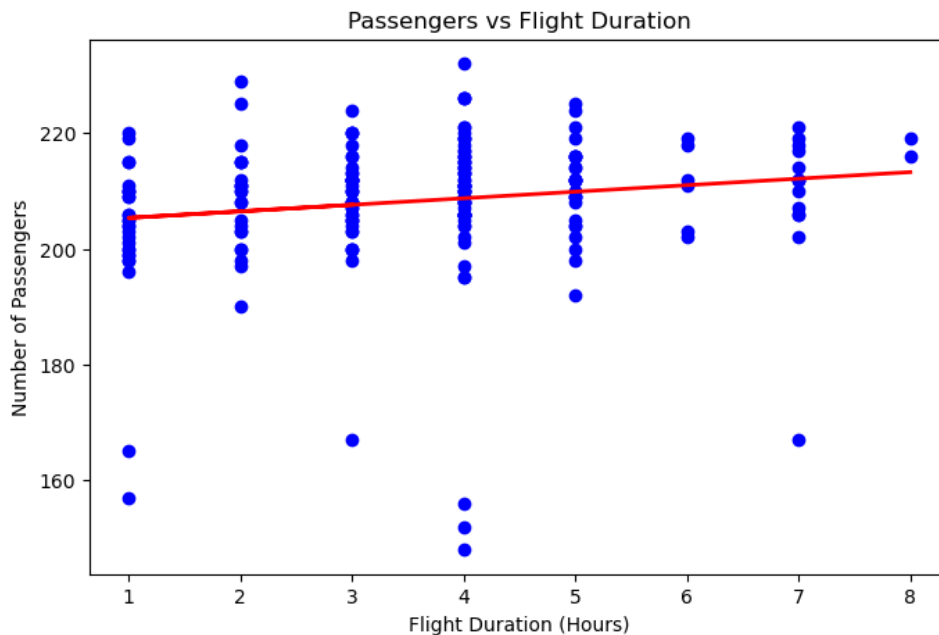


Figure 18: Passengers vs Flight Duration with Regression Line

7.5 Regression Equation

The regression equation is:

$$y = 204.21 + 1.13x$$

Where,

- y = Number of Passengers
- x = Flight Duration (Hours)

7.6 Interpretation

The correlation coefficient between flight duration and number of passengers is 0.159, which indicates a very weak positive relationship between the two variables.

The scatter plot shows that passenger numbers remain relatively stable across different flight durations. Most flights carry around 200 to 220 passengers regardless of the number of hours.

The regression line has a slight upward slope, suggesting that passenger numbers increase slightly as flight duration increases. According to the regression equation, each additional hour of flight duration increases the number of passengers by approximately 1.13 passengers on average.

However, the relationship is very weak because the points are widely scattered and the correlation value is low. Therefore, flight duration does not strongly affect the number of passengers.

Question no 8

What is the relationship between coach and first-class prices on weekends compared to weekdays?

8 Solution:

8.1 Python Code

```
# =====
# Question 8
# Relationship Between Coach and First-Class Prices
# on Weekends vs Weekdays
# =====
# Average Prices by Weekend

avg_prices = flight.groupby(
    "weekend"
)[["coach_price", "firstclass_price"]].mean()

print("Average Prices by Weekend:")
print(avg_prices)

# Boxplot for Coach Prices
plt.figure(figsize=(8,5))
flight.boxplot(
    column="coach_price",
    by="weekend",
    patch_artist=True,
    boxprops=dict(facecolor="skyblue"))
plt.title("Coach Prices on Weekends vs Weekdays")
plt.suptitle("")
plt.xlabel("Weekend")
plt.ylabel("Coach Price")
plt.show()

# Boxplot for First-Class Prices
plt.figure(figsize=(8,5))
flight.boxplot(
    column="firstclass_price",
    by="weekend",
    patch_artist=True,
    boxprops=dict(facecolor="orange"))
plt.title("First-Class Prices on Weekends vs Weekdays")
plt.suptitle("")
plt.xlabel("Weekend")
plt.ylabel("First-Class Price")
plt.show()
```



```

# Correlation on Weekdays

weekday_data = flight[flight["weekend"] == "No"]

weekday_corr = weekday_data["coach_price"].corr(
    weekday_data["firstclass_price"]
)

print("\nCorrelation on Weekdays =", weekday_corr)

# Correlation on Weekends

weekend_data = flight[flight["weekend"] == "Yes"]

weekend_corr = weekend_data["coach_price"].corr(
    weekend_data["firstclass_price"]
)

print("Correlation on Weekends =", weekend_corr)

```

8.2 Results

Average Prices by Weekend:

	coach_price	firstclass_price
No	328.11	1256.17
Yes	406.86	1555.65

Correlation on Weekdays = 0.6197982932381213

Correlation on Weekends = 0.4923473967994085

8.3 Coach Prices on Weekends vs Weekdays



Figure 19: Coach Prices on Weekends vs Weekdays

8.4 First-Class Prices on Weekends vs Weekdays

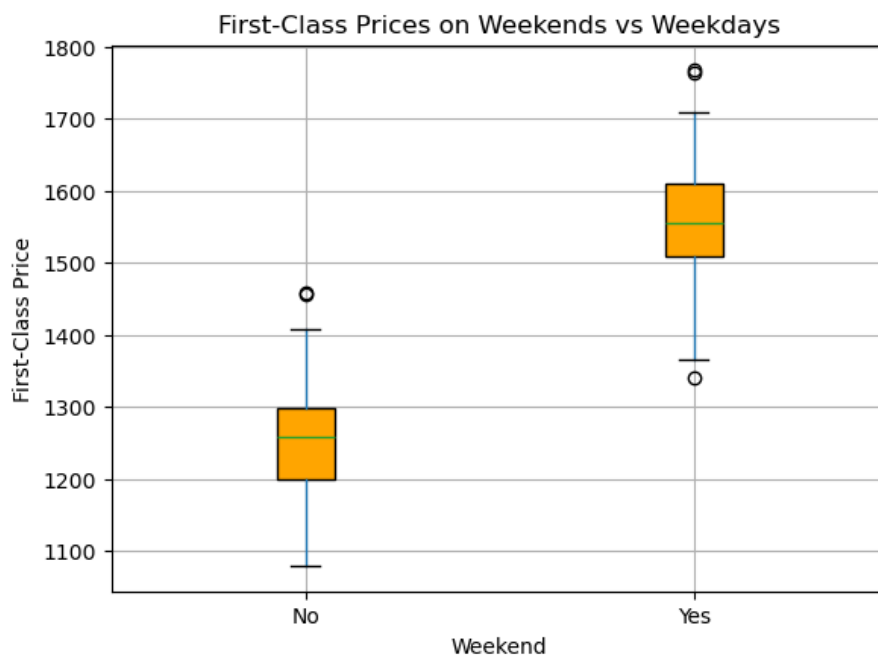


Figure 20: First-Class Prices on Weekends vs Weekdays

8.5 Interpretation

The analysis shows that both coach ticket prices and first-class ticket prices are generally higher on weekends compared to weekdays.

The average coach ticket price on weekdays is approximately \$328.11, while on weekends it increases to about \$406.86.

Similarly, the average first-class ticket price rises from approximately \$1256.17 on weekdays to about \$1555.65 on weekends.

The boxplots clearly show that the median prices for both coach and first-class tickets are higher during weekends. Weekend flights also show a wider spread in prices, indicating greater price variation.

The correlation between coach and first-class prices on weekdays is 0.620, which indicates a moderately strong positive relationship. On weekends, the correlation decreases slightly to 0.492, indicating a moderate positive relationship.

This means that although higher coach prices are generally associated with higher first-class prices in both cases, the relationship is somewhat stronger during weekdays than weekends.

Question no 9

How do coach prices differ for redeyes and non-redeyes on each day of the week?

9 Solution:

9.1 Python Code

```
# =====  
# Question 9  
# Coach Prices for Redeye and Non-Redeye Flights  
# by Day of Week  
# =====  
# Average Coach Prices by Day and Redeye  
avg_prices = flight.groupby(  
    ["day_of_week", "redeye"]  
    ["coach_price"].mean()  
print("Average Coach Prices by Day and Redeye:")  
print(avg_prices)  
# Boxplot for Redeye vs Non-Redeye  
plt.figure(figsize=(8,5))  
flight.boxplot(  
    column="coach_price",  
    by="redeye",  
    patch_artist=True,  
    boxprops=dict(facecolor="skyblue"))  
plt.title("Coach Prices for Redeye vs Non-Redeye Flights")  
plt.suptitle("")  
plt.xlabel("Redeye Flight")  
plt.ylabel("Coach Price")  
plt.show()
```

```

# Interaction Plot

import seaborn as sns

plt.figure(figsize=(10,5))

sns.lineplot(
    data=flight,
    x="day_of_week",
    y="coach_price",
    hue="redeye",
    marker="o"
)

plt.title("Average Coach Prices by Day and Redeye")
plt.xlabel("Day of Week")
plt.ylabel("Average Coach Price")

plt.show()

```

9.2 Results

Average Coach Prices by Day and Redeye:

Friday	No	404.16
Monday	No	354.84
Monday	Yes	296.62
Saturday	No	419.58
Saturday	Yes	298.42
Sunday	No	407.29
Sunday	Yes	229.15
Thursday	No	305.42
Tuesday	No	330.54
Tuesday	Yes	414.39
Wednesday	No	287.68

9.3 Coach Prices for Redeye vs Non-Redebye Flights

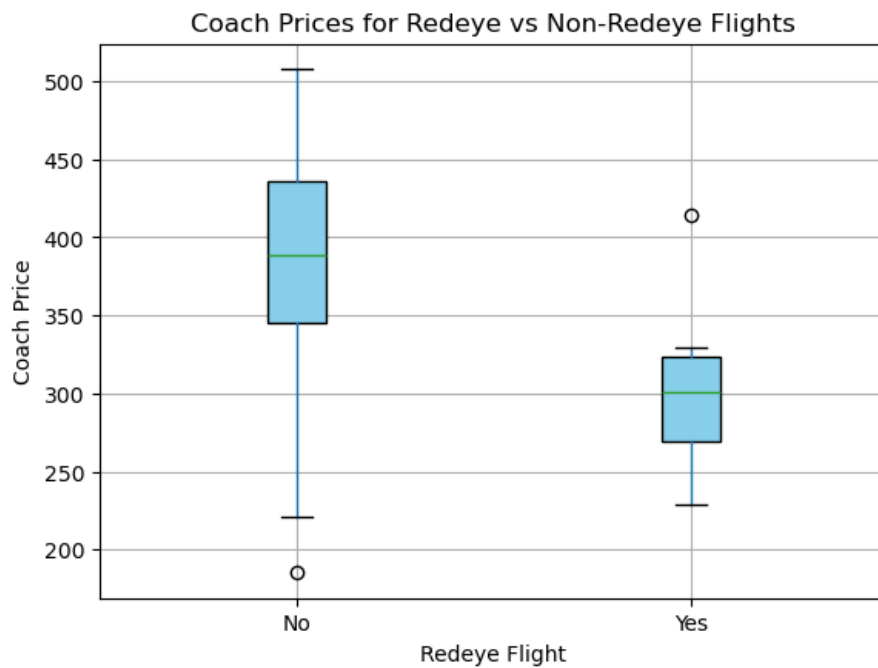


Figure 21: Coach Prices for Redeye vs Non-Redebye Flights

9.4 Average Coach Prices by Day and Redeye

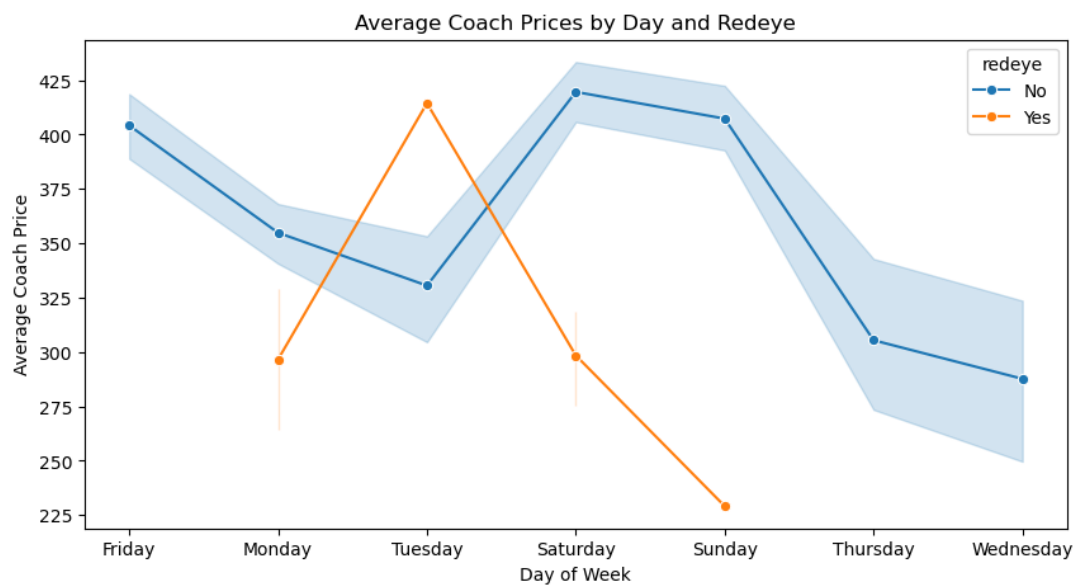


Figure 22: Average Coach Prices by Day and Redeye

9.5 Interpretation

The analysis shows that non-redeye flights generally have higher coach ticket prices than redeye flights on most days of the week.

The boxplot indicates that non-redeye flights have a higher median coach price and a wider variation in ticket prices compared to redeye flights.

From the interaction plot, non-redeye flights show relatively high prices on Friday, Saturday, and Sunday, with average prices above \$400. In contrast, redeye flights usually have lower prices, especially on Sunday where the average coach price is approximately \$229.15.

An exception appears on Tuesday, where redeye flights have a higher average coach price (\$414.39) than non-redeye flights (\$330.54).

Overall, redeye flights tend to be cheaper than non-redeye flights for most days of the week. This suggests that overnight flights are generally offered at lower prices, possibly to attract more passengers.

Question no 10

Perform the analysis for appropriate variables: summary statistics, visualization, hypothesis testing, correlation analysis, regression analysis, and logistic regression.

10 Solution:

10.1 Python Code

```
# =====
# Question 10
# Overall Statistical Analysis
# =====
# Summary Statistics
print(flight[[
    "coach_price",
    "firstclass_price",
    "miles",
    "passengers",
    "delay",
    "hours"
]].describe())
# Correlation Matrix
correlation_matrix = flight[[
    "coach_price",
    "firstclass_price",
    "miles",
    "passengers",
    "delay",
    "hours"
]].corr()
print(correlation_matrix)
```

```

# Independent t-test
from scipy.stats import ttest_ind
weekend_prices = flight[
    flight["weekend"] == "Yes"
]["coach_price"]

weekday_prices = flight[
    flight["weekend"] == "No"
]["coach_price"]

t_stat, p_value = ttest_ind(
    weekend_prices,
    weekday_prices)

print("t-statistic =", t_stat)
print("p-value =", p_value)

# Linear Regression
from sklearn.linear_model import LinearRegression

X = flight[["coach_price"]]
y = flight["firstclass_price"]

model = LinearRegression()

model.fit(X, y)

print(
    "y =",
    round(model.intercept_, 2),
    "+",
    round(model.coef_[0], 2),
    "x"
)

# Logistic Regression
from sklearn.linear_model import LogisticRegression
flight["red-eye_binary"] = flight["red-eye"].map({
    "Yes": 1,
    "No": 0})
X_log = flight[[
    "coach_price",
    "hours",
    "delay"]]
y_log = flight["red-eye_binary"]
log_model = LogisticRegression()
log_model.fit(X_log, y_log)
print("Logistic Regression Completed Successfully")

```

10.2 Distribution of Coach Ticket Prices

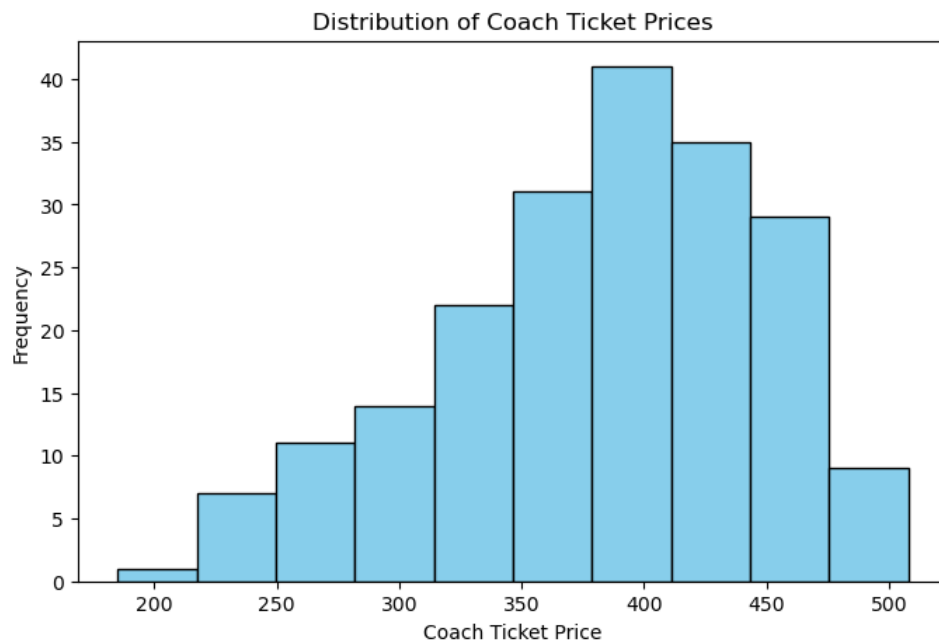


Figure 23: Distribution of Coach Ticket Prices

10.3 Boxplot of Coach Ticket Prices

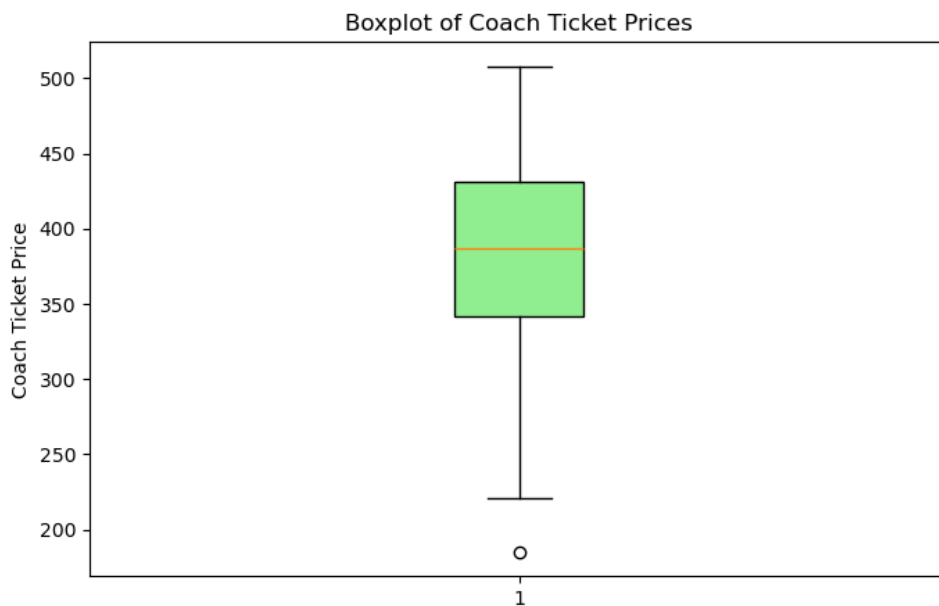


Figure 24: Boxplot of Coach Ticket Prices

10.4 Flights by Day of Week

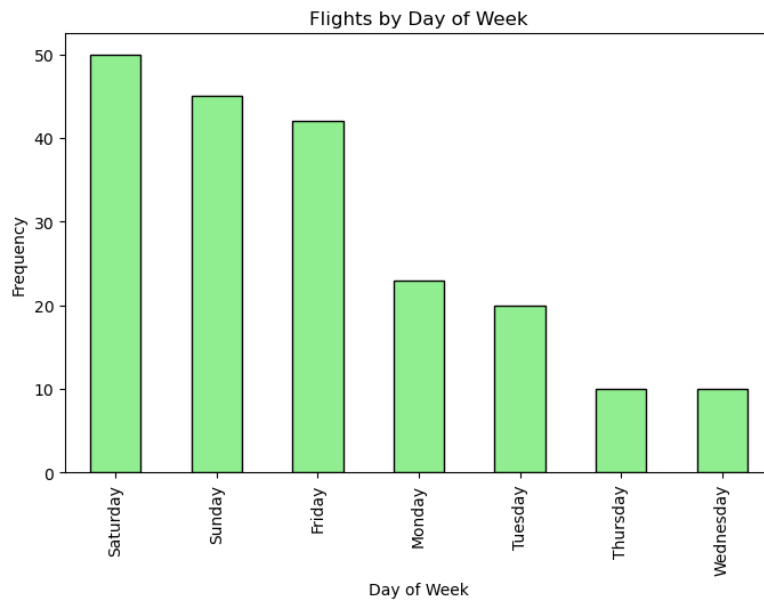


Figure 25: Flights by Day of Week

10.5 Correlation Heatmap

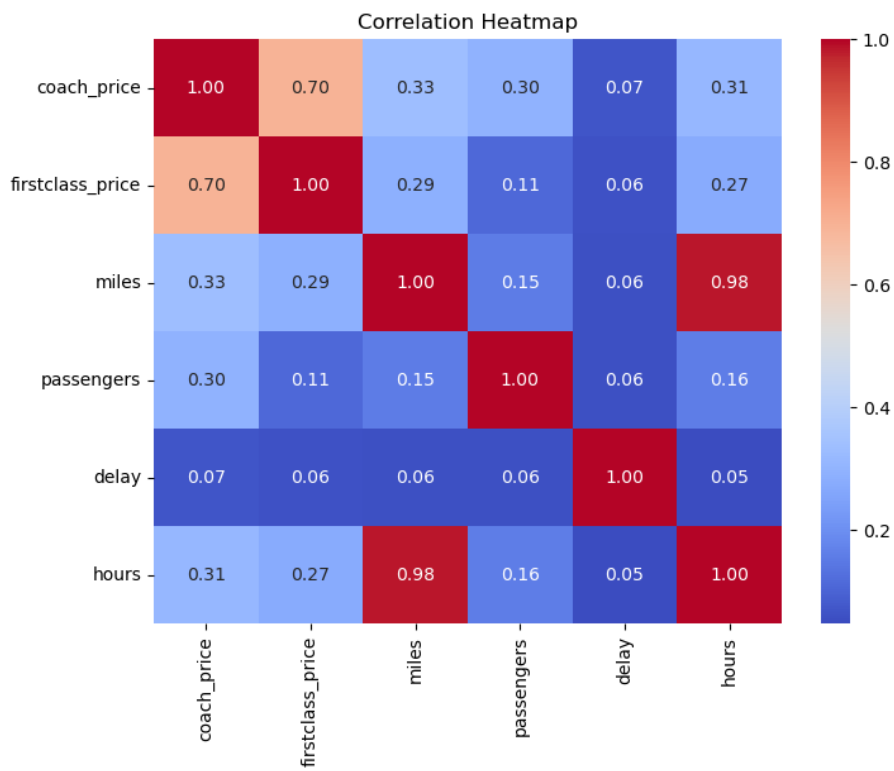


Figure 26: Correlation Heatmap

10.6 Results

Independent t-test

t-statistic = 9.338656970746412

p-value = 2.0607479155760792e-17

Linear Regression Equation:

$$y = 823.53 + 1.67x$$

Logistic Regression Completed Successfully

10.7 Correlation Matrix

	coach_price	firstclass_price
coach_price	1.000000	0.698050
firstclass_price	0.698050	1.000000
miles	0.332224	0.288250
passengers	0.296786	0.107536
delay	0.069226	0.059584
hours	0.310325	0.273756

10.8 Interpretation

The histogram shows that most coach ticket prices are concentrated between \$300 and \$450, indicating a moderate distribution of ticket prices.

The boxplot indicates that the median coach ticket price is around \$390, with a few lower-price outliers.

The bar chart shows that most flights occur on Saturday and Sunday, while Wednesday and Thursday have the fewest flights.

The correlation heatmap reveals a strong positive relationship between miles and hours (0.98), meaning longer distances require longer flight durations. Coach prices also have a strong positive correlation with first-class prices (0.70).

The independent t-test result has a very small p-value, indicating a statistically significant difference between weekend and weekday coach ticket prices.

The linear regression equation:

$$y = 823.53 + 1.67x$$

shows that first-class prices increase as coach ticket prices increase.

Finally, logistic regression was successfully performed to predict redeye flights using coach price, hours, and delay variables.